

Class 8: Mini-Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of breast mass.

Data Import

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"
# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)
wisc.df
```

```
head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
84348301	0.14250	0.28390	0.2414	0.10520	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
84348301	0.05963	0.009208	14.91	26.50	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0	0.1622	0.6656	
842517	158.80	1956.0	0.1238	0.1866	
84300903	152.50	1709.0	0.1444	0.4245	
84348301	98.87	567.7	0.2098	0.8663	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
84348301	0.6869	0.2575	0.6638		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				
84348301	0.17300				

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B    M
357 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us here:

```
colnames(wisc.data)
```

```
[1] "radius_mean"      "texture_mean"
[3] "perimeter_mean"   "area_mean"
[5] "smoothness_mean"  "compactness_mean"
[7] "concavity_mean"   "concave.points_mean"
[9] "symmetry_mean"    "fractal_dimension_mean"
[11] "radius_se"        "texture_se"
[13] "perimeter_se"     "area_se"
[15] "smoothness_se"    "compactness_se"
[17] "concavity_se"     "concave.points_se"
[19] "symmetry_se"      "fractal_dimension_se"
[21] "radius_worst"     "texture_worst"
[23] "perimeter_worst"  "area_worst"
[25] "smoothness_worst" "compactness_worst"
[27] "concavity_worst"  "concave.points_worst"
[29] "symmetry_worst"   "fractal_dimension_worst"
```

```
length(grep("_mean",colnames(wisc.data),value=TRUE))
```

```
[1] 10
```

Principal Component Analysis

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```
wisc.pr <- prcomp(wisc.data,scale=T)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

The proportion of the variance that is captured by PC1 is 0.4427 which is 44.27%.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

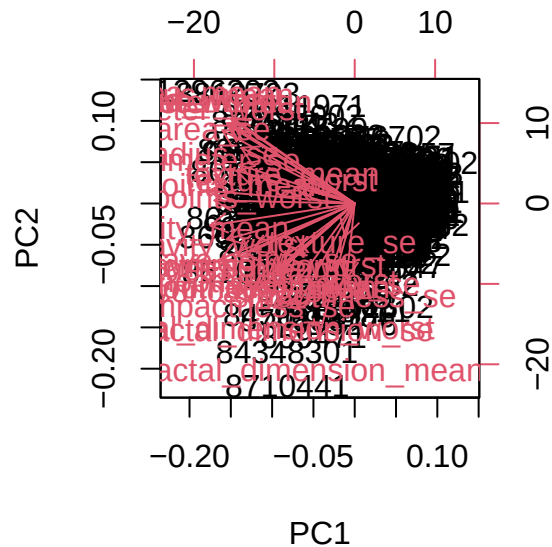
Three principal components are required to describe at least 70% of the original variance. Three is the first principal component that is 70% or more.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Seven principal components are required to describe at least 90% of the original variance in the data because the seventh principal component is the first one to have 90% or more.

Let's see the "PC score plot"

```
biplot(wisc.pr)
```

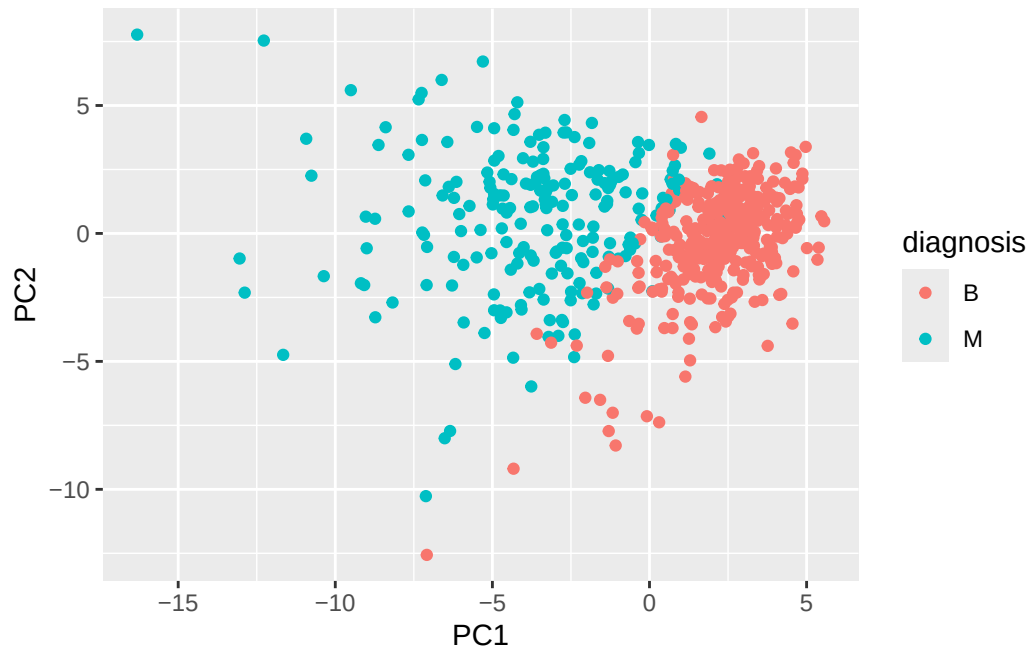


Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

This plot is not easy to understand. It is extremely cluttered and hard to see anything that is going on. The thing that stands out the most to me in this plot is the giant black circle in the middle.

```
library(ggplot2)

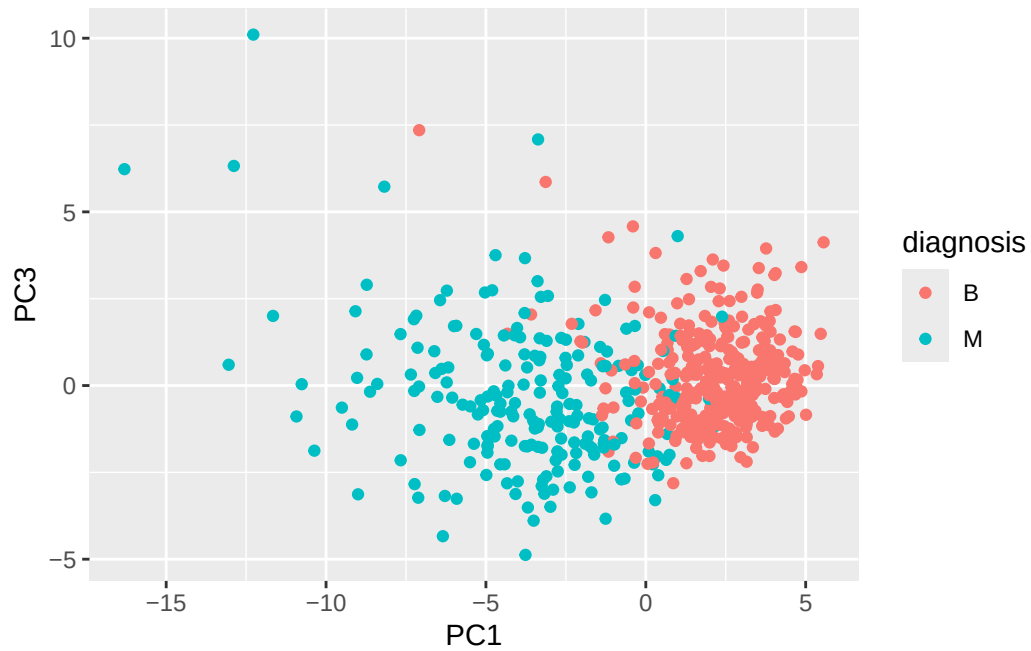
ggplot(wisc.pr$x)+aes(PC1,PC2,col=diagnosis)+geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

The plots look relatively similar, however I can tell that the plots are different by looking at the isolated points on the edges. Overall, the plot looks like it has shifted down.

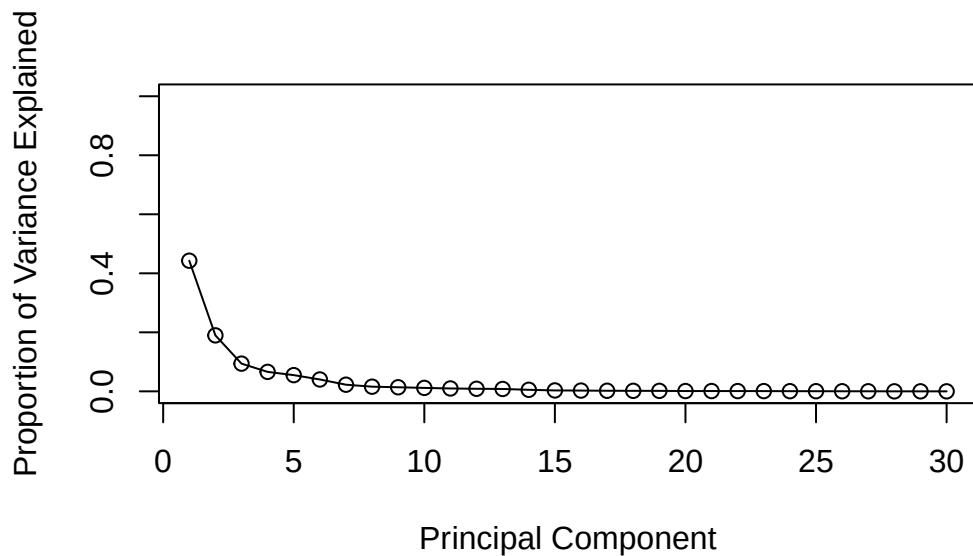
```
ggplot(wisc.pr$x) + aes(PC1, PC3, col = diagnosis) + geom_point()
```



```
pr.var <- wisc.pr$sdev^2  
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
pve <- pr.var / sum(pr.var)  
  
plot(pve, xlab = "Principal Component",  
      ylab = "Proportion of Variance Explained",  
      ylim = c(0, 1), type = "o")
```



Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
sort(abs(wisc.pr$rotation[, 1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se
0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se

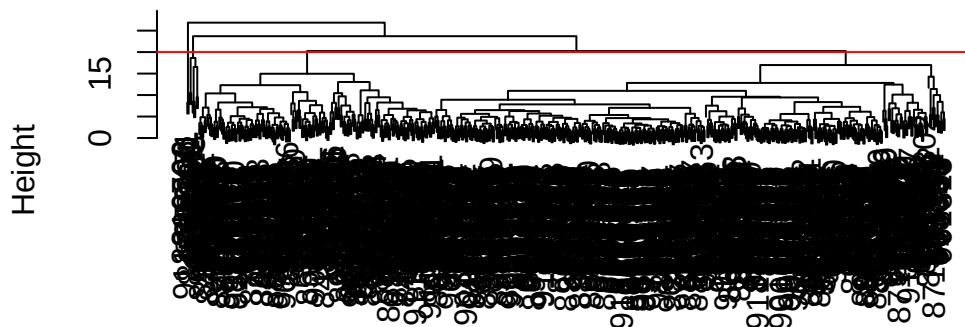
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

No, there is not any features with larger contributions than principal component 1.

Hierarchial Clustering

```
wisc.hclust<-hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
abline(h=20,col="red")
```

Cluster Dendrogram



```
dist(scale(wisc.data))
hclust (*, "complete")
```

Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

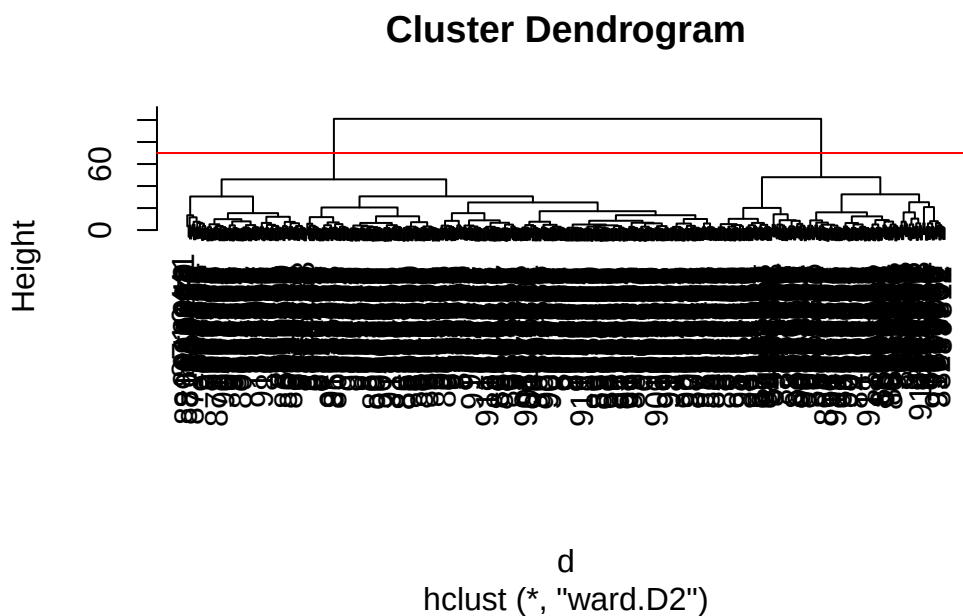
Using the plot, the height at which there is 4 clusters seems to be around 16-18.

Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

The table method gives more favorable results as its much clearer to interpret and provides clear output.

Combining Methods

```
d<- dist(wisc.pr$x[,1:4])
wisc.pr.hclust<-hclust(d,method="ward.D2")
plot(wisc.pr.hclust)
abline(h=70,col="red")
```



```
grps<-cutree(wisc.pr.hclust,h=70)
table(grps)
```

```
grps
  1  2
171 398
```

How does this clustering `grps()` correspond to the expert diagnosis?

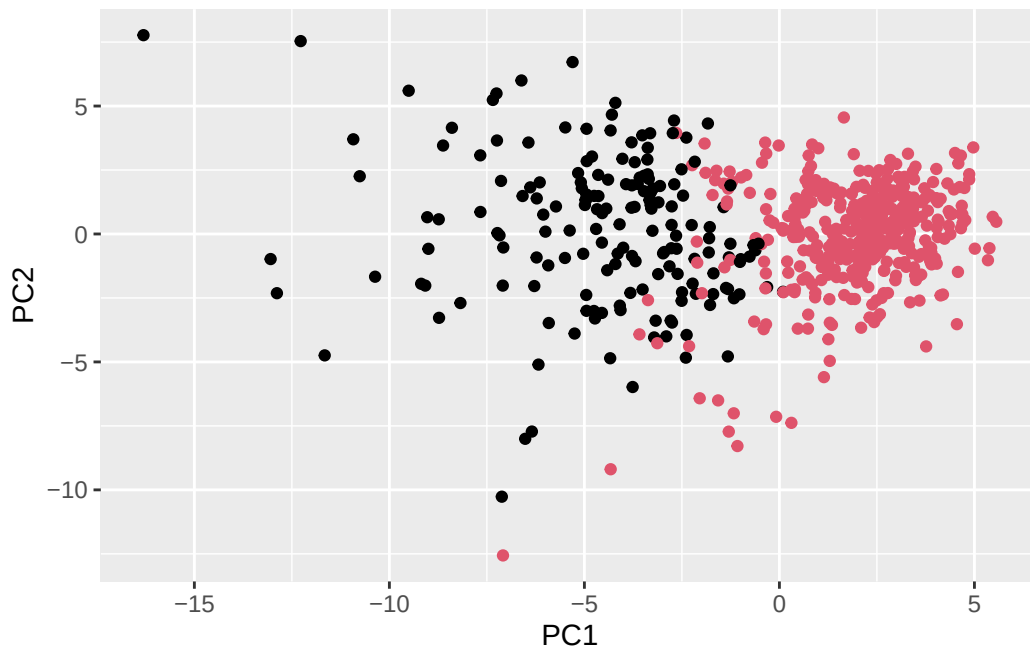
```
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

```
table(diagnosis,grps)
```

```
      grps
diagnosis  1   2
  B      6 351
  M     165  47
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



Q13. How well does the newly created `hclust` model with two clusters separate out the two “M” and “B” diagnoses?

The new model does a decent job at modeling the two separate clusters, there is no key so differentiating between “M” and “B” is unclear.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B   M
1         6 165
2        351  47
```

The new model is more accurate than the older model because it gives us a better output and interpretation of the results.

Prediction

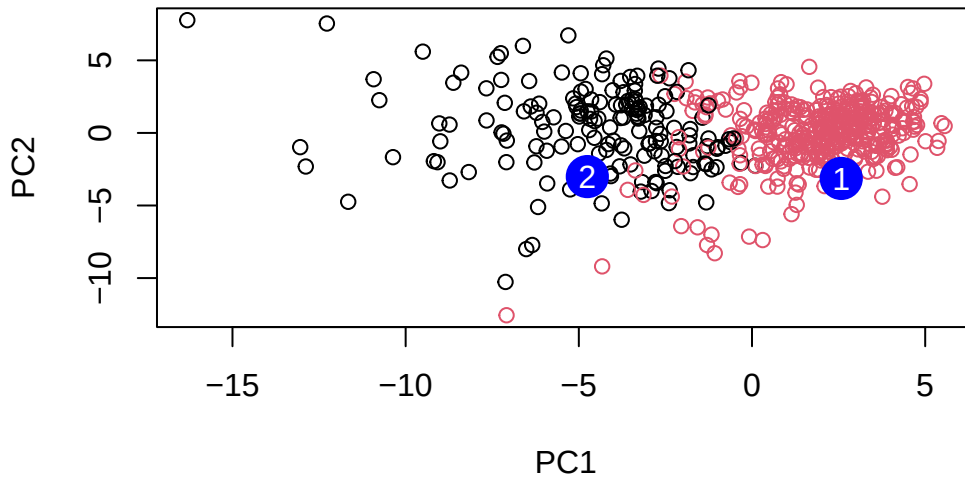
```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```

      PC1      PC2      PC3      PC4      PC5      PC6      PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
      PC8      PC9      PC10     PC11     PC12     PC13     PC14
[1,] -0.2307350  0.1029569 -0.9272861  0.3411457  0.375921  0.1610764  1.187882
[2,] -0.3307423  0.5281896 -0.4855301  0.7173233 -1.185917  0.5893856  0.303029
      PC15     PC16     PC17     PC18     PC19     PC20
[1,]  0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,]  0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
      PC21     PC22     PC23     PC24     PC25     PC26
[1,]  0.1228233  0.09358453  0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776  0.01732146  0.06316631 -0.2338618 -0.20755948 -0.009833238
      PC27     PC28     PC29     PC30
```

```
[1,] 0.220199544 -0.02946023 -0.015620933 0.005269029  
[2,] -0.001134152 0.09638361 0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)  
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)  
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize patient 2 as the models say the black cluster is malignant, while the red cluster is benign.